Stabilization of a two-wheel mobile robot

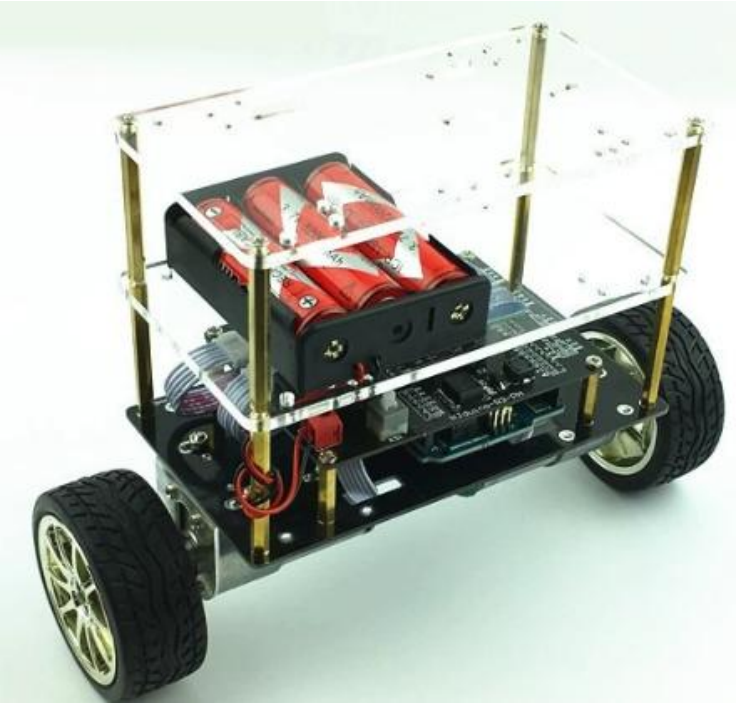
RBE203TC

Group assignment

Objective

- Stability control of two-wheel mobile robot
  - PID control
  - Active balancing during movement
  - Wireless control
  - 3 tasks

- General rules for tasks:
  - The robot must maintain balance throughout each task. If the robot falls over, the trial is considered a failure.
  - Each team is allowed three trials per task.
  - The highest score obtained from the three trials will be used as the final score for each task.

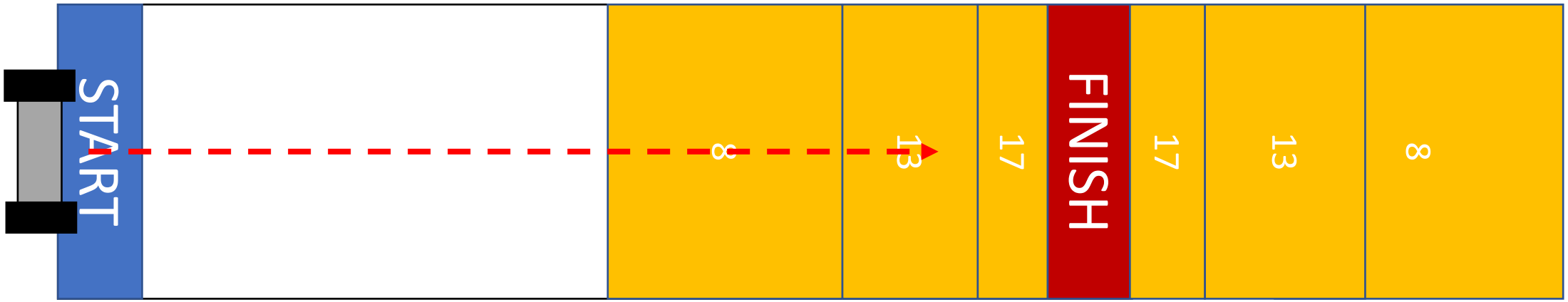


Tasks to accomplish

- High-speed movement with stability (20p):
  - Achieve the highest possible movement speed while maintaining balance, directional accuracy, and stable motion.
- Stability on tilted terrain (40p):
  - Maintain balance and controlled movement on a laterally tilted surface (side slope) without falling over.
- Stability on inclined and declined terrain (40p):
  - Maintain balance and controlled movement while ascending and descending an inclined surface without losing control and falling over.

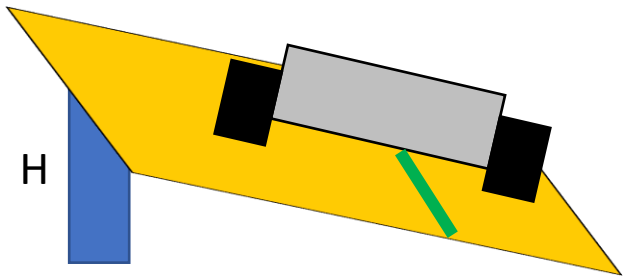
High-speed movement with stability

- Travel 5 meters within 10 seconds
  - Scoring:
    - The score is determined by the robot's distance from the finish line after 5 seconds. The closer the robot is to the finish line, the higher the score.
  - Failure:
    - If the robot falls over or fails to reach the 8-point scoring region, a score of 5 points is assigned.

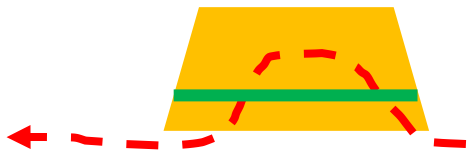


Stability on tilted terrain

- Travel across a laterally tilted platform while maintaining stability.
- Two wheel must be on the platform during travelling
- Tilt conditions:
  - Raising one side to five different heights (H): 2, 4, 6, 8, and 10 cm
- Scoring:
  - Points are awarded based on the maximum tilt height successfully completed.
    - Failure: 10p
    - 2 cm: 20p (one wheel: 12p)
    - 4 cm: 25p (one wheel: 15p)
    - 6 cm: 30p (one wheel: 18p)
    - 8 cm: 35p (one wheel: 21p)
    - 10 cm: 40p (one wheel: 24p)



Two wheels are all on the platform

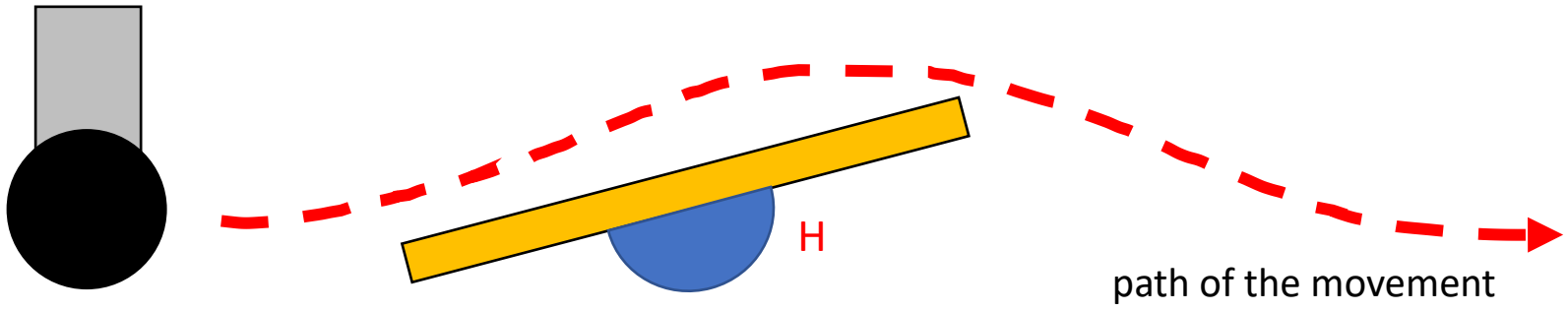


path of the movement

* For one wheel to be considered over the platform, it must cross the designated boundary line.

Stability on inclined and declined terrain

- Travel across a see-saw platform while maintaining balance and stability.
- Platform Conditions:
  - central section's height (H) is adjusted to five levels: 2, 4, 6, 8, and 10 cm.
- Scoring:
  - Points are awarded based on the maximum platform height successfully completed without falling over.
    - Failure: 10p
    - 2 cm: 20p
    - 4 cm: 25p
    - 6 cm: 30p
    - 8 cm: 35p
    - 10 cm: 40p



Post-test cleanup and equipment return

- Workspace cleanup:
  - Students must clean and organize their desks, chairs, and surrounding work areas after completing the test.
- Equipment return:
  - Students must place the mobile robot back into its box and return it to the TA.
- TA Verification:
  - The TA will inspect the workspace and confirm that the robot has been properly returned.
- Penalty:
  - Failure to complete the cleanup and equipment return procedures will result in a 50% deduction from the test score.

Procedure

1. **Robot Preparation**
   - Collect your mobile robot from Room B2021.
   - Complete the assembly of the robot and charge its battery.
   - Download the provided code from Learning Mall (Week 6).
   - Test the robot, modify the code as necessary, and upload the updated code to the robot.
   - Each task might require different setting
2. **Performance Testing**
   - Once your group is ready, approach the instructor to begin each task.
   - Each group is allowed a maximum of three trials per task.
   - Approach to the instructor when the robot is ready before each trial.
   - Complete all three performance evaluation tasks.
3. **Cleanup and Equipment Return**
   - Clean and organize your desks, chairs, and surrounding work areas.
   - Place the mobile robot back into its original storage box and hand it to the TA.
   - Obtain confirmation from the TA that the workspace has been properly cleaned and the robot has been returned.
   - Return the robot box to its original storage location in Room B2021.
4. **Attendance Confirmation**
   - After completing all procedures, approach to the instructor for attendance confirmation.